

GWAS, dose-0 consistency, growth rate and IC50 on the DH4148 reference

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GWAS, dose-0 consistency, growth rate and IC50 on the DH4148 reference

Generated 2026-10-09. Phenotypes: Chromium, Copper and Lead screens (Iron and Zinc are held out, D-56). Genotypes: popgen callset on the DH4148 assembly (GCA_058775505.1). Panel: 126 haploid *R. mucilaginosa* strains (D-49). All numbers come from `report/tables/*.csv`. Scripts: `scripts/01` to `scripts/12`.

Summary

1. **The 126-strain panel is 14 clonal lineages.** Strains that differ by at most 2,000 SNPs form 14 groups (sizes 42, 34, 16, 6, 6, 6, 6, 3, 2 and five single strains). Within a lineage the median difference is 557 SNPs. Between lineages it is at least 48,641 SNPs. The 5-SNP de-cloning of D-49 does not remove this structure. The effective sample size for a genome-wide scan is about 14, not 126.
2. **Genome-wide hits are lineage markers.** With the relatedness matrix only, 64 loci pass the Bonferroni threshold (unadjusted phenotypes) and 44 loci pass it (run-adjusted phenotypes). **All of them are SNPs that do not vary inside any lineage.** They mark which lineage a strain belongs to. They cannot be mapped to a gene.
3. **Within lineages, nothing passes the threshold.** With run adjustment and lineage covariates the scans find 0 Bonferroni loci in 20 traits. 17 SNPs reach the suggestive level, against about 94 expected by chance. The 12 suggestive loci are listed for reference (Table 13). They are hypotheses, not results.
4. **Lineage explains part of some traits.** Intraclass correlation by lineage: Cr relative area at the top dose 0.29, Cr area AUC 0.28, Cu a* AUC 0.26, dose-0 ln area 0.24, dose-0 a* 0.23, Cu relative area at the top dose 0.20. Pb traits and all growth-rate traits are at most 0.10. With 14 lineages these estimates are uncertain.
5. **Non-additive variance is not detectable.** The strains are haploid, so there is no dominance. For epistasis, the additive-by-additive variance component is not significant in any of 20 traits (smallest $p = 0.095$). A scan of about 65,000 SNP pairs per trait found no pair below $p = 1e-5$ (about 0.65 expected by chance).
6. **Dose-0 (YPD) controls are consistent for a*, less so for size and not for growth rate.** Strain explains 38% of dose-0 a* variance across the three screens, and the screen itself explains about 0%. For ln area, strain explains 18% and the screen 17%. For the maximum growth rate, strain explains 3%.
7. **Growth rate and IC50.** Relative growth rate falls with dose in Cr (0.41 of control at dose 1.2) and Pb (0.12 at dose 30). Cu barely changes it (0.93 at dose 30). IC50 of relative area could be fitted for almost all strains in Cr and Pb. In Cu 138 of 317 strains are right-censored, because relative area stays at or above 0.5 at the top dose.

1. Data

- **Phenotypes** come from the DuckDB `heavy_metal_measurement` table with strain, species and ploidy from the `strain_info` view (D-58). a* is `ColorLab_a*GeoMedian`. The largest object per well and image is used. The late window and the 2,000 px colony cutoff follow the a* report (`analysis/carotenoid_stress_vs_baseline`).

- **Growth rate** of a well is the largest slope of ln area against time over windows of 5 consecutive images. A window must start at 300 px or more and span at least 4 h. A well needs at least 6 images.
- **Relative traits** compare a strain with itself at dose 0: area AUC (relative area against dose scaled 0 to 1), relative area at the top dose, relative growth-rate AUC, a* change at the top dose and a* AUC.
- **IC50** is fitted for relative area with a two-parameter log-logistic curve. It is reported only when the curve crosses 0.5 inside the tested dose range.
- **GWAS traits** are 20 per strain (Table 1). Each is rank-normalised before the scan. Copper IC50 is not used (46% of panel strains are censored).
- **Genotypes.** Biallelic SNPs from popgen's `rmuc_core` file, kept at minor allele frequency 0.05 or more and at most 10% missing genotypes inside the panel: 244,956 SNPs. After collapsing identical genotype patterns there are 7,199 distinct patterns (the effective number of tests for the lineage-free scans). With lineage covariates there are 2,940.
- **Annotation.** snpEff effects come from the VCF. Gene IDs and product names come from the DH4148 GFF. No GO or KEGG annotation exists for DH4148 yet.

trait	count	mean	std	min	50%	max
d0_a	125.000	0.896	2.060	-5.236	1.321	6.291
d0_lnA	125.000	-0.004	0.109	-0.542	0.011	0.251
d0_rgr	125.000	0.000	0.002	-0.005	0.000	0.004
Cr_da_top	125.000	-12.880	5.731	-20.941	-14.248	0.299
Cr_a_auc	125.000	-1.123	1.567	-4.887	-1.263	1.930
Cr_relarea_top	125.000	0.152	0.416	0.038	0.084	3.575
Cr_relarea_auc	125.000	0.591	0.067	0.450	0.577	0.897
Cr_relrgr_auc	124.000	0.880	0.135	0.662	0.847	1.271
Cr_logIC50	123.000	-0.471	0.105	-0.791	-0.477	-0.071
Cu_da_top	125.000	-15.411	3.007	-21.478	-15.299	-3.941
Cu_a_auc	125.000	-6.333	2.197	-13.938	-6.017	1.586
Cu_relarea_top	125.000	0.547	0.236	0.105	0.489	1.240
Cu_relarea_auc	125.000	1.006	0.213	0.549	0.988	2.377
Cu_relrgr_auc	125.000	1.117	0.166	0.845	1.100	1.681
Pb_da_top	123.000	-11.111	7.199	-19.034	-13.645	10.347
Pb_a_auc	123.000	-0.822	3.121	-6.732	-1.587	7.057
Pb_relarea_top	123.000	0.128	0.109	0.046	0.085	0.585
Pb_relarea_auc	123.000	0.542	0.125	0.336	0.506	0.947
Pb_relrgr_auc	122.000	0.734	0.148	0.438	0.708	1.171
Pb_logIC50	119.000	2.580	0.114	2.102	2.591	2.819

Table 1. GWAS traits in the 126-strain panel (raw values; 119-125 strains per trait).

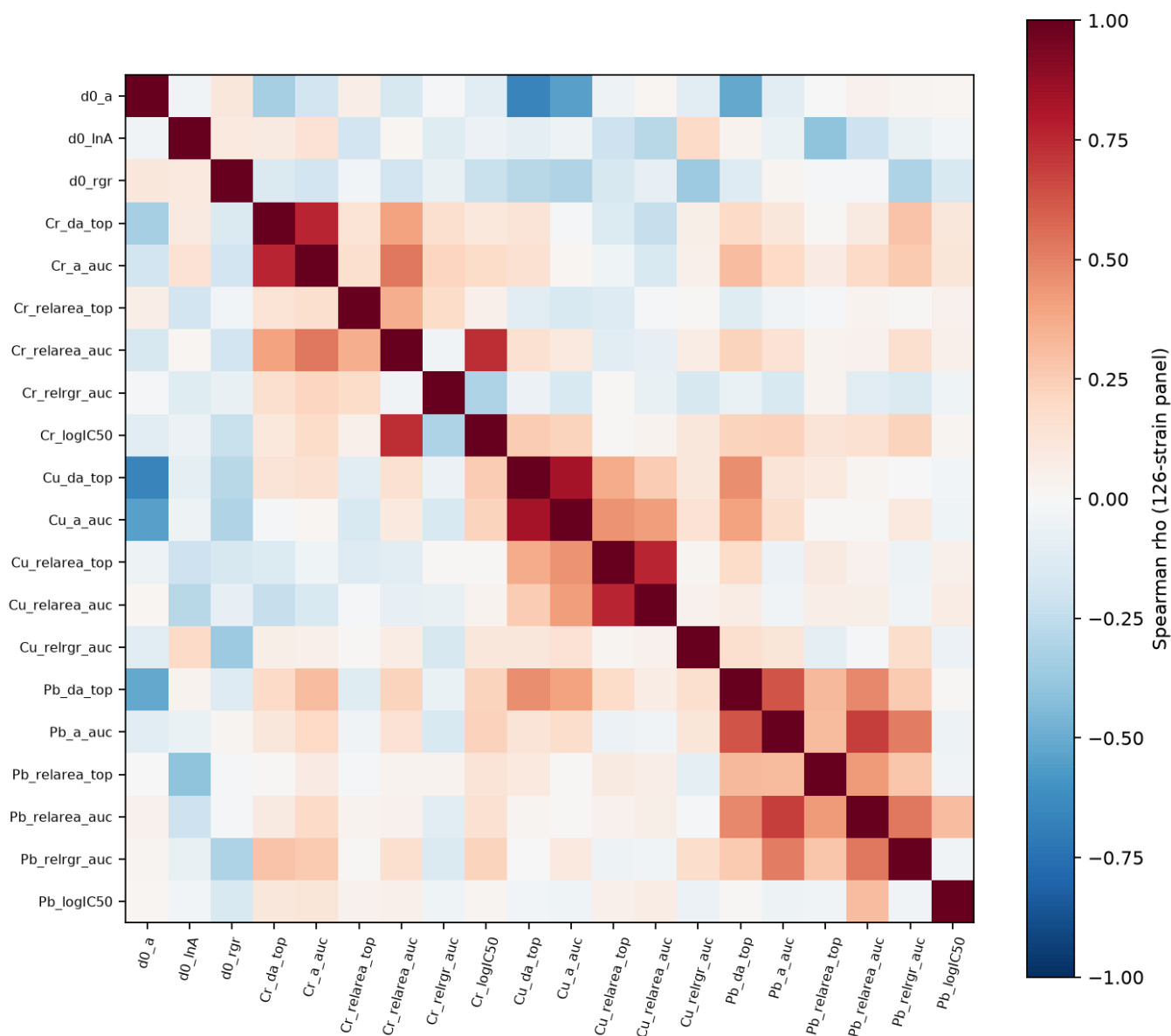


Figure 1. Spearman correlation among the 20 traits (panel strains).

2. Dose-0 (YPD) consistency across experiments

Dose-0 wells of the three screens are pooled. Model: $\text{trait} \sim 1 + (1 \mid \text{strain}) + (1 \mid \text{screen}) + (1 \mid \text{run}) + (1 \mid \text{plate})$. Wells are replicate wells on dose-0 plates. Fe and Zn dose-0 plates exist but are partial, so the main model uses Cr, Cu and Pb. A version with all five screens is in `report/tables/dose0_variance_components.csv`.

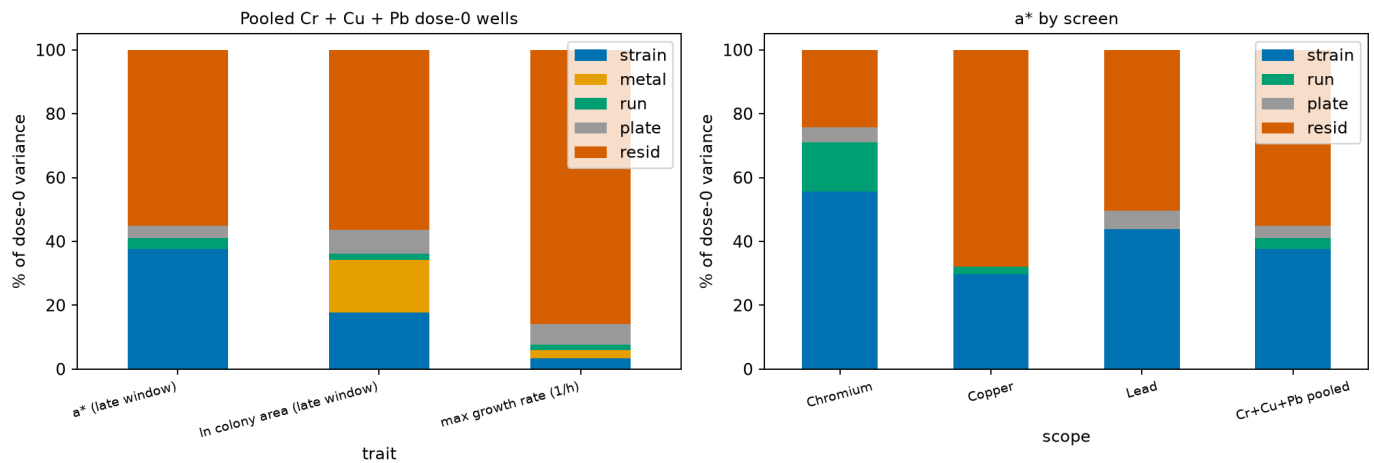


Figure 2. Shares of dose-0 variance: strain, screen (metal), run, plate and residual.

scope	trait	n_wells	n_strains	share_strain	share_metal	share_run	share_plate	share_resid
Cr+Cu+Pb pooled	a	3282	319	0.38	0.00	0.03	0.04	0.55
Chromium	a	1188	306	0.56		0.15	0.05	0.24
Copper	a	1116	319	0.30		0.02	0.00	0.68
Lead	a	978	297	0.44		0.00	0.06	0.50
Cr+Cu+Pb pooled, GWAS panel strains	a	1371	125	0.28	0.00	0.01	0.08	0.62
Cr+Cu+Pb pooled	lnA	3282	319	0.18	0.16	0.02	0.07	0.56
Chromium	lnA	1188	306	0.58		0.09	0.04	0.30
Copper	lnA	1116	319	0.13		0.02	0.06	0.80
Lead	lnA	978	297	0.29		0.00	0.13	0.58
Cr+Cu+Pb pooled, GWAS panel strains	lnA	1371	125	0.13	0.20	0.01	0.11	0.55
Cr+Cu+Pb pooled	rgr	3412	320	0.03	0.03	0.02	0.06	0.86
Chromium	rgr	1194	309	0.07		0.10	0.02	0.81
Copper	rgr	1141	319	0.04		0.00	0.06	0.90
Lead	rgr	1077	307	0.14		0.00	0.08	0.78
Cr+Cu+Pb pooled, GWAS panel strains	rgr	1405	125	0.02	0.04	0.03	0.06	0.85

Table 2. Variance shares of dose-0 wells. *share_metal* is empty for single-screen models. Fits marked singular in the table have a variance component at zero.

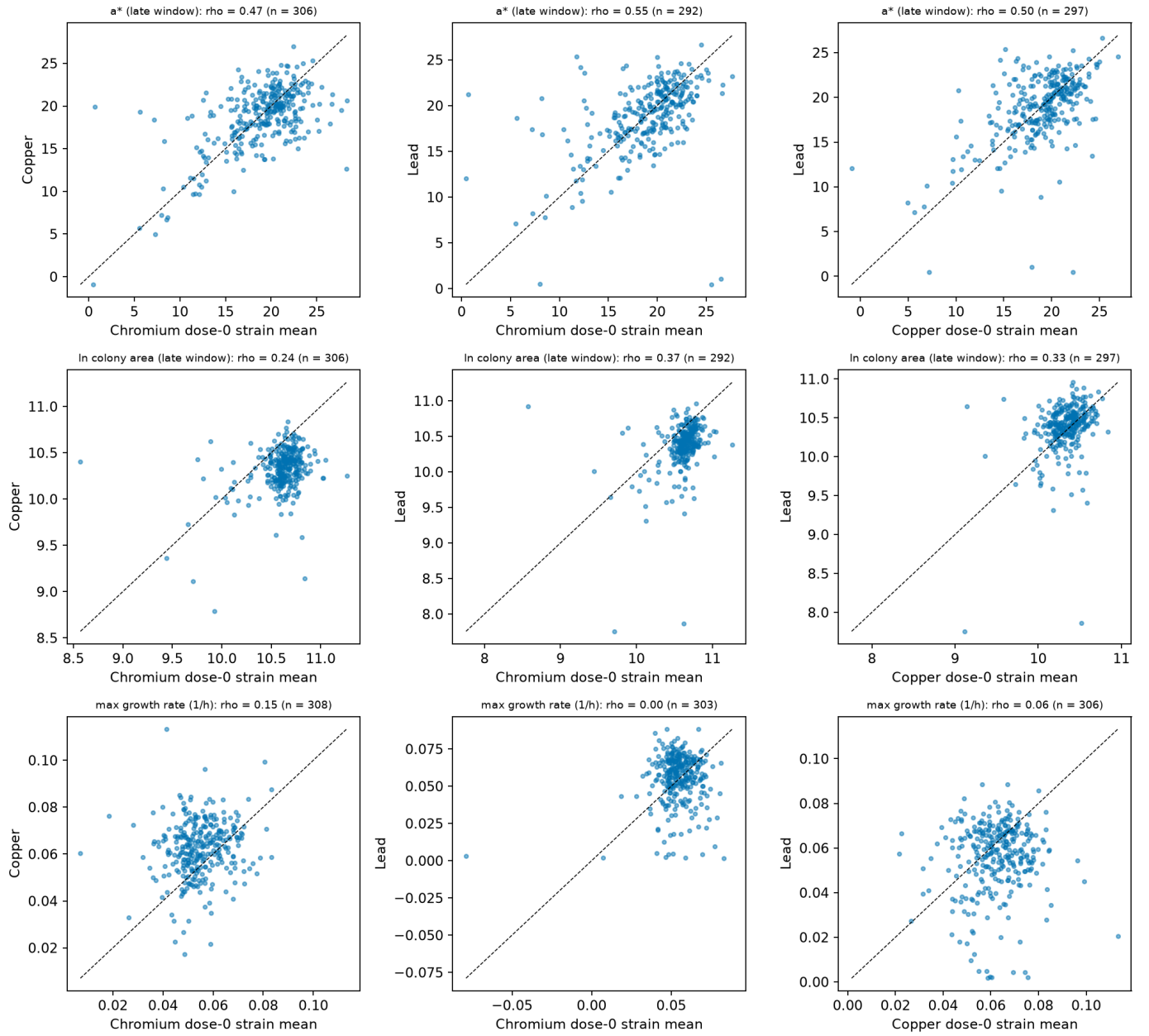


Figure 3. Dose-0 strain means in one screen against another (dashed = equal values).

trait	screen_1	screen_2	n_strains	spearman	pearson
a	Chromium	Copper	306	0.47	0.59
a	Chromium	Lead	292	0.55	0.48
a	Copper	Lead	297	0.50	0.55
lnA	Chromium	Copper	306	0.24	0.32
lnA	Chromium	Lead	292	0.37	0.32
lnA	Copper	Lead	297	0.33	0.33
rgr	Chromium	Copper	308	0.15	0.14
rgr	Chromium	Lead	303	0.00	0.11
rgr	Copper	Lead	306	0.06	0.06

Table 3. Agreement of dose-0 strain means between screens.

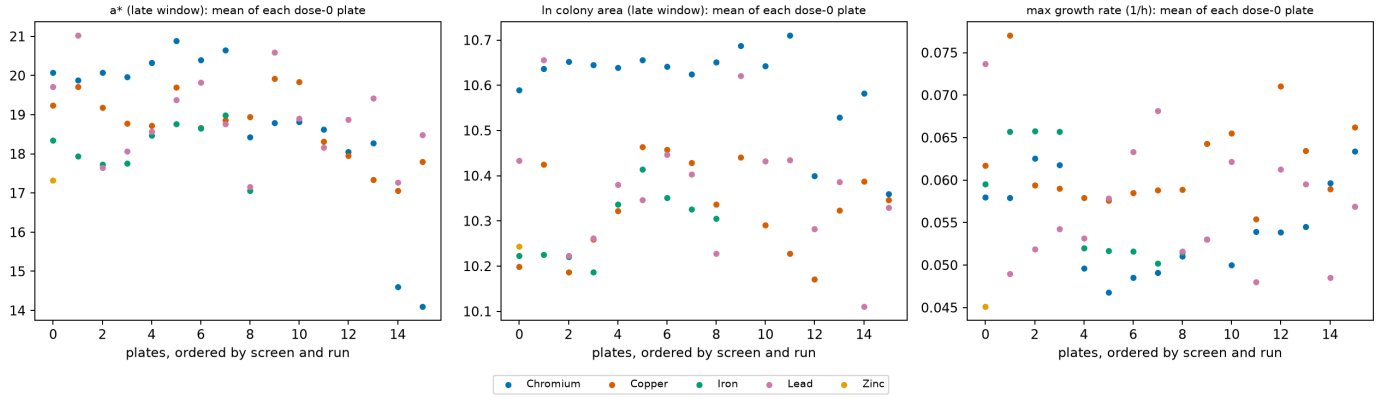


Figure 4. Mean of each dose-0 plate, by screen.

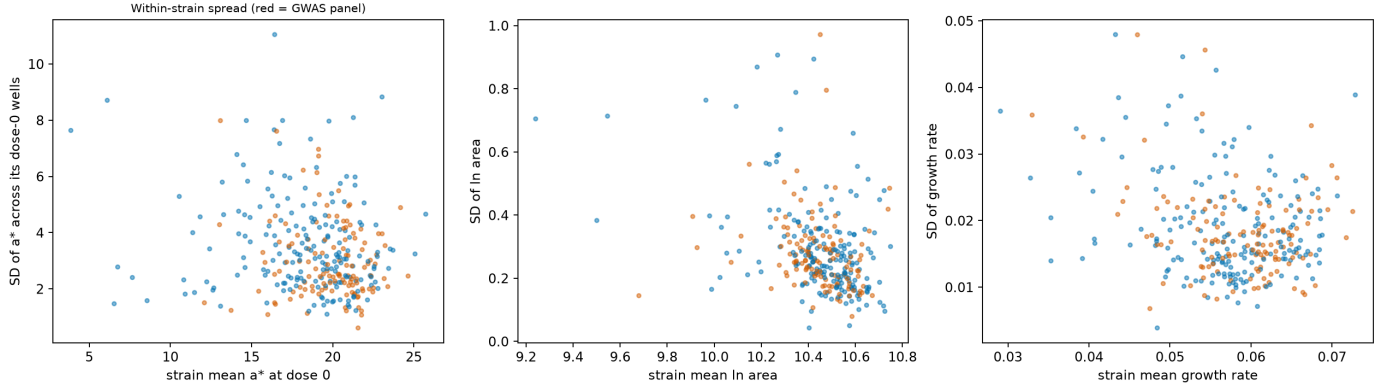


Figure 5. Spread of a^* , ln area and growth rate across the dose-0 wells of each strain (red = GWAS panel).*

sample_name	species	n_wells	a_mean	a_sd	a_sd_z
TFCN_25-333Y-2	R. sp_clade_I	11	16.43	11.05	5.88
TFCN_25-333D-1	R. sp_clade_I	8	23.04	8.83	4.26
DBVPG_6660	R. frigidialcoholis	10	6.11	8.71	4.18
TFCN_86A-8	R. sphaerocarpa	11	21.28	8.10	3.73
DBVPG_4952	R. mucilaginosa	7	16.91	7.99	3.65
DBVPG_6083	R. graminis	10	14.68	7.99	3.65
DBVPG_3236	R. mucilaginosa	11	13.08	7.99	3.65
TFCN_86A-5	R. sphaerocarpa	11	19.77	7.97	3.64
TFCN_86C-10	R. sphaerocarpa	7	16.40	7.66	3.41
NRRL_Y-7192	R. sphaerocarpa	11	3.87	7.64	3.40
DBVPG_4203	R. mucilaginosa	11	16.55	7.61	3.38
DBVPG_3769	R. dairenensis	12	18.64	7.33	3.17

Table 4. Strains whose dose-0 wells vary most in a^* (robust z of the SD across wells; strains with at least 6 wells).

Reading.

- **a^* at dose 0 is a reproducible strain trait.** Strain explains 38% of the variance in the pooled model, and the screen explains about 0%. Strain means agree between screens (Spearman 0.47 to 0.55). Run and plate explain 3% and 4%. Half of the variance (55%) is replicate-well noise.
- **Colony size at dose 0 differs between screens.** The screen explains 17% of ln area variance, against 18% for strain. Plate means of ln area are 10.60 (Cr), 10.33 (Cu) and 10.37 (Pb). Strain means agree only weakly between screens (0.24 to 0.38).
- **Growth rate is not a reproducible strain trait here.** Strain explains 3% of its variance and between-screen correlations are 0.00 to 0.15. It is not suitable for mapping.
- **Cr dose-0 plates vary most in a^* .** In the Cr-only model run explains 15% of the variance, against 2% in Cu and 0% in Pb. The plate-mean SD of a^* is 1.98 in Cr, 0.87 in Cu and 1.10 in Pb.

- **Within-strain spread.** The median SD of a^* across a strain's dose-0 wells is 2.98. 13 of 312 strains have a robust z above 3 (Table 4). Spread alone does not show mixed cultures. The identity columns of `strain_curation.csv` (`identity_flag_categories`) should be checked for these strains; this has not been done here.
- **Panel strains.** In the panel strain subset the strain share of a^* is 28% (all strains 38%).

3. Growth rate and IC50

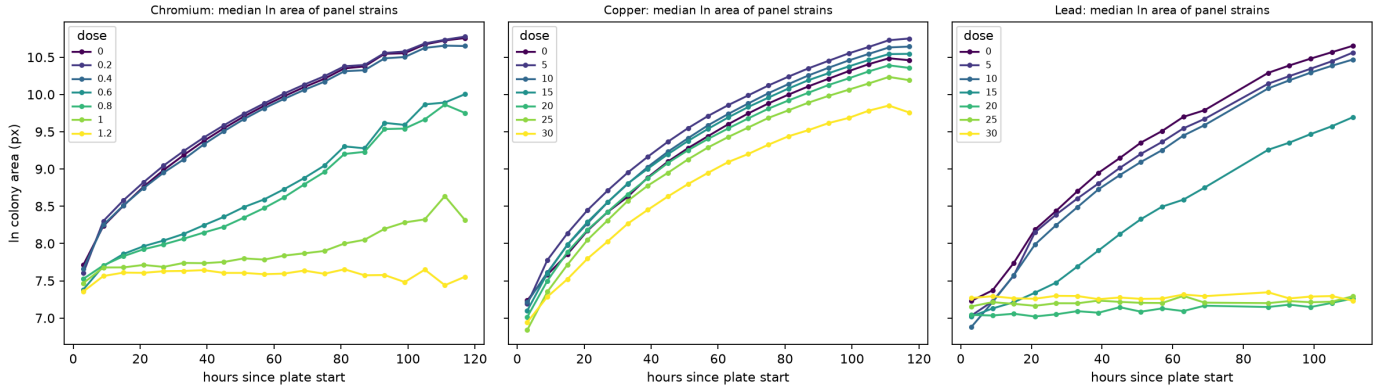


Figure 6. Median $\ln$ colony area of panel strains against time, by dose.

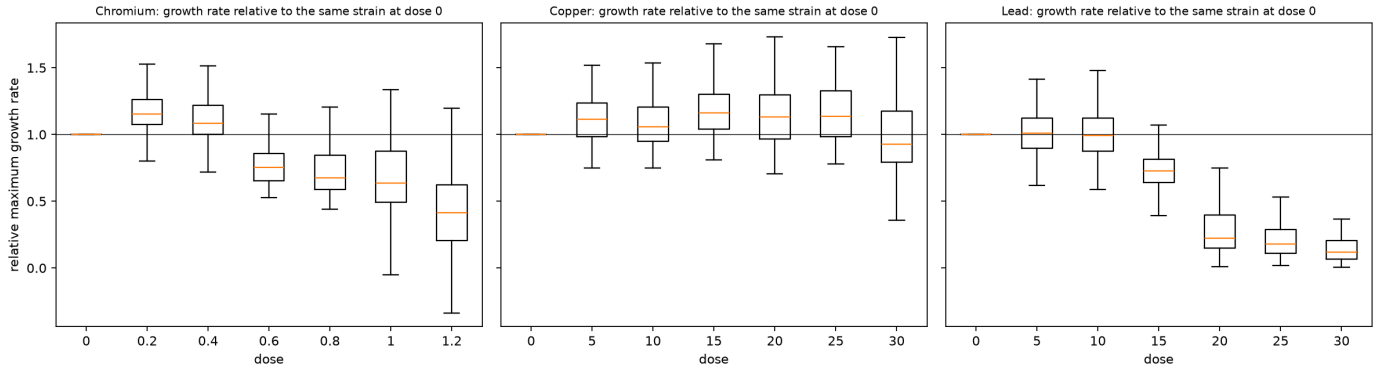


Figure 7. Maximum growth rate of each strain relative to the same strain at dose 0, by dose (panel strains).

Metal	dose	n_strains	median_rel_rgr	q25	q75
Chromium	0.00	125	1.00	1.00	1.00
Chromium	0.20	125	1.15	1.07	1.26
Chromium	0.40	125	1.08	1.00	1.22
Chromium	0.60	125	0.75	0.65	0.86
Chromium	0.80	125	0.68	0.59	0.84
Chromium	1.00	124	0.64	0.49	0.88
Chromium	1.20	102	0.41	0.20	0.62
Copper	0.00	125	1.00	1.00	1.00
Copper	5.00	125	1.11	0.98	1.23
Copper	10.00	125	1.06	0.95	1.20
Copper	15.00	125	1.16	1.04	1.30
Copper	20.00	125	1.13	0.97	1.30
Copper	25.00	125	1.14	0.98	1.33
Copper	30.00	125	0.93	0.79	1.17
Lead	0.00	123	1.00	1.00	1.00
Lead	5.00	123	1.01	0.90	1.12
Lead	10.00	123	0.99	0.88	1.12
Lead	15.00	123	0.73	0.64	0.81
Lead	20.00	67	0.22	0.15	0.40
Lead	25.00	76	0.18	0.11	0.29

Metal	dose	n_strains	median_rel_rgr	q25	q75
Lead	30.00	81	0.12	0.07	0.20

Table 5. Relative maximum growth rate by dose (median and quartiles across strains). Pb at doses 20 to 30 has only 67 to 81 strains with enough images for a growth rate.

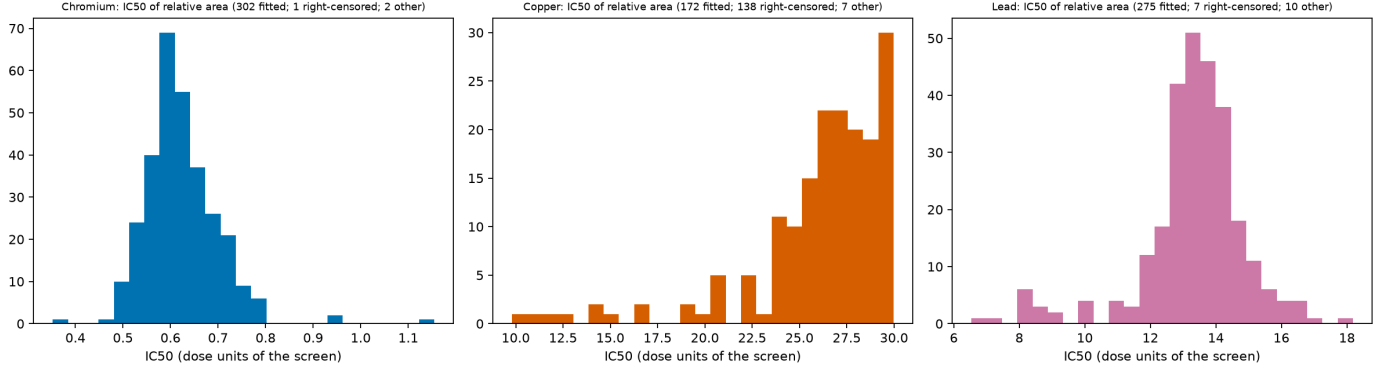


Figure 8. IC50 of relative area (all strains).

Metal	ok	outside_range	right_censored
Chromium	302	2	1
Copper	172	7	138
Lead	275	10	7

Table 6. IC50 fit status by metal (all strains).

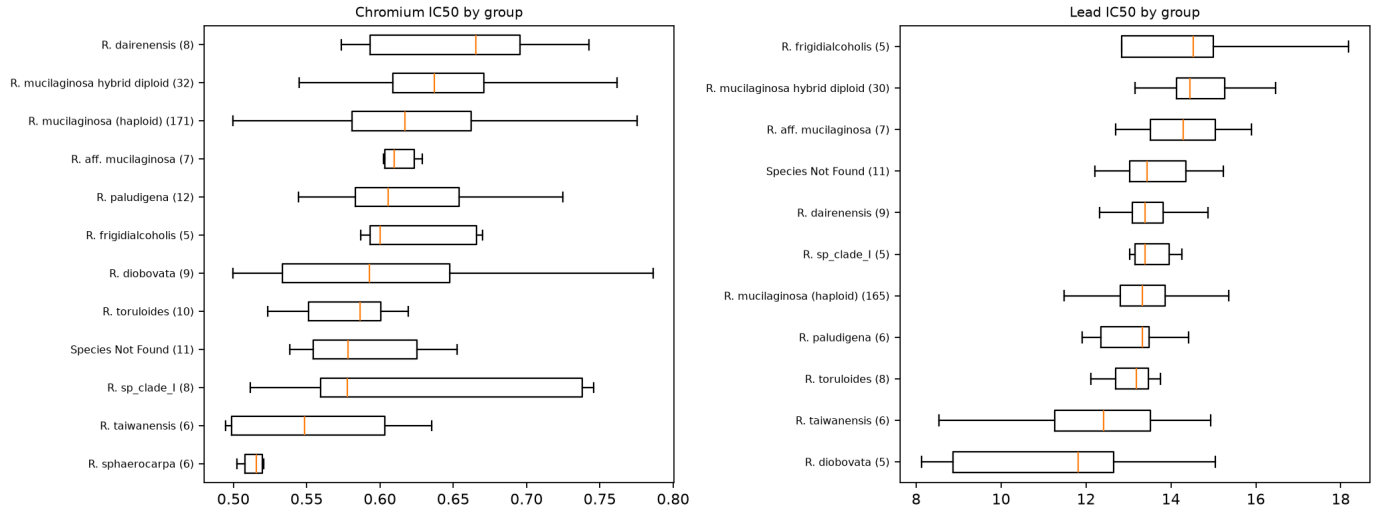


Figure 9. IC50 by species and ploidy group (groups with at least 5 fitted strains; Cr and Pb).

Metal	group	n	median_ic50	q25	q75
Chromium	R. aff. mucilaginosa	7	0.61	0.60	0.62
Chromium	R. dairenensis	8	0.67	0.59	0.70
Chromium	R. diobovata	9	0.59	0.53	0.65
Chromium	R. frigidialcoholis	5	0.60	0.59	0.67
Chromium	R. mucilaginosa (haploid)	171	0.62	0.58	0.66
Chromium	R. mucilaginosa hybrid diploid	32	0.64	0.61	0.67
Chromium	R. paludigena	12	0.61	0.58	0.65

Metal	group	n	median_ic50	q25	q75
Chromium	R. sp_clade_I	8	0.58	0.56	0.74
Chromium	R. sphaerocarpa	6	0.52	0.51	0.52
Chromium	R. taiwanensis	6	0.55	0.50	0.60
Chromium	R. toruloides	10	0.59	0.55	0.60
Chromium	Species Not Found	11	0.58	0.55	0.63
Copper	R. aff. mucilaginosa	5	27.16	25.58	29.59
Copper	R. dairenensis	9	27.92	22.19	28.50
Copper	R. diobovata	6	26.66	24.28	28.28
Copper	R. mucilaginosa (haploid)	86	27.49	25.28	29.10
Copper	R. mucilaginosa hybrid diploid	20	26.97	26.01	27.54
Copper	R. paludigena	11	25.03	23.90	28.37
Copper	R. sp_clade_I	5	27.80	25.53	28.23
Copper	R. toruloides	7	26.54	26.06	28.59
Lead	R. aff. mucilaginosa	7	14.28	13.52	15.04
Lead	R. dairenensis	9	13.39	13.10	13.82
Lead	R. diobovata	5	11.82	8.86	12.65
Lead	R. frigidialcoholis	5	14.53	12.84	15.00
Lead	R. mucilaginosa (haploid)	165	13.33	12.80	13.86
Lead	R. mucilaginosa hybrid diploid	30	14.44	14.13	15.27
Lead	R. paludigena	6	13.32	12.35	13.48
Lead	R. sp_clade_I	5	13.39	13.15	13.96
Lead	R. taiwanensis	6	12.41	11.26	13.52
Lead	R. toruloides	8	13.18	12.70	13.47
Lead	Species Not Found	11	13.43	13.03	14.36

Table 7. IC50 (dose units of the screen) by group. Copper IC50 values are close to the top dose (30) and half of the strains are censored, so Copper IC50 is coarse.

Reading.

- **Cr:** growth rate rises at low doses (1.15 at 0.2) and falls to 0.64 at dose 1.0 and 0.41 at dose 1.2. Median IC50 is about 0.6 dose units for pure haploid *R. mucilaginosa* (IQR 0.58 to 0.66).
- **Cu:** growth rate stays at 0.93 to 1.16 of control through dose 30. Relative area falls only at the top doses. Most Cu IC50 values are outside the tested range.
- **Pb:** growth rate is unchanged to dose 10, 0.73 at dose 15, and 0.12 at dose 30. Median IC50 is about 13.3 for pure haploid strains.
- **Group differences.** *R. sphaerocarpa* and *R. taiwanensis* have the lowest Cr IC50 (0.52 and 0.55, 6 strains each). Hybrid diploids have a higher Pb IC50 (14.4) than pure haploids (13.3). Group sizes are small (5 to 12 strains), so these are descriptive.
- **IC50 against other traits.** Cr IC50 correlates with Cr area AUC (rho 0.74) but not with the a* change (0.10). Pb IC50 correlates weakly with Pb area AUC (0.31).

4. Genetic structure: 14 lineages

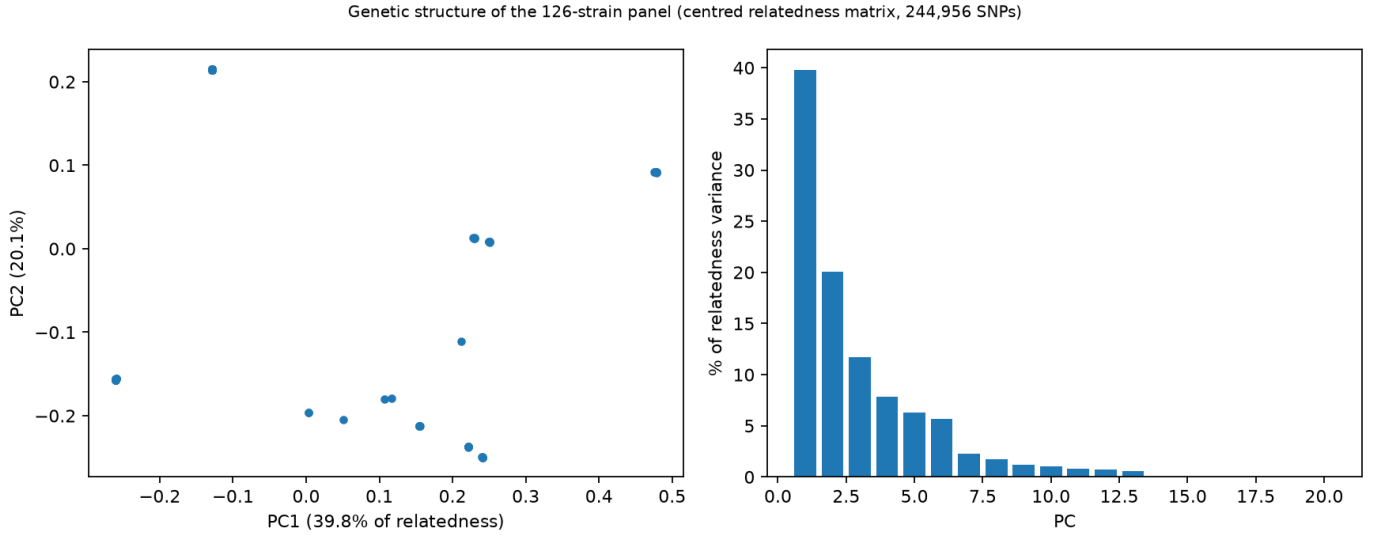


Figure 10. First two principal components of the centred relatedness matrix and the variance of the first 20 components. Each visible point is a cluster of near-identical strains.

max_snp_differences_within_group	n_groups	largest_groups
5	126	1, 1, 1, 1, 1, 1
50	93	10, 7, 5, 4, 4, 3
500	35	41, 14, 8, 6, 6, 6
2000	14	42, 34, 16, 6, 6, 6
5000	14	42, 34, 16, 6, 6, 6
20000	14	42, 34, 16, 6, 6, 6
50000	13	76, 16, 6, 6, 6, 6

Table 8a. Number of single-linkage groups of panel strains by the largest allowed SNP difference. The count is 14 for any cut from 2,000 to 20,000 differences.

lineage	n_strains	example_strains
L01	42	TFCN_98A-5, TFCN_102C-2, TFCN_105A-2, TFCN_17-334Y-2
L02	34	TFCN_107A-6, TFCN_102D-1, TFCN_4M-1-2, TFCN_2M-1-4
L03	16	DBVPG_3775, DBVPG_3776, DBVPG_3777, DBVPG_4920
L04	6	DBVPG_3381, DBVPG_3382, DBVPG_3383, DBVPG_3384
L05	6	DBVPG_3983, DBVPG_3984, DBVPG_3044, DBVPG_3090
L06	6	DBVPG_5228, DBVPG_5229, DBVPG_5235, DBVPG_10882
L07	6	DBVPG_4408, DBVPG_4376, DBVPG_5757, DBVPG_5758
L08	3	DBVPG_4945, DBVPG_4947, DBVPG_10842
L09	2	DBVPG_4958, DBVPG_10937
L10	1	TFCN_17-329Y-4
L11	1	DBVPG_4378
L12	1	DBVPG_4618
L13	1	TFCN_25-334D-3
L14	1	DBVPG_4377

Table 8b. Lineage sizes (lineage 1 is the largest).

- The first two components explain 40% and 20% of the relatedness. The panel has visible clusters, not a continuum.
- Lineage and run are associated in all three screens (chi-square $p < 0.001$). Strains of one lineage were often on the same runs. Removing run effects therefore removes part of the lineage signal as well.

Lineage as a source of trait variance

trait	ICC raw	ICC run-adjusted	p (run-adjusted, ANOVA)
Cr_a_auc	0.32	0.14	0.01
Cr_da_top	0.14	0.07	0.11
Cr_logIC50	0.14	0.12	0.02
Cr_relarea_auc	0.27	0.28	0.00
Cr_relarea_top	0.32	0.29	0.00
Cr_relrgr_auc	0.11	0.07	0.12
Cu_a_auc	0.25	0.26	0.00
Cu_da_top	0.17	0.17	0.00
Cu_relarea_auc	0.02	0.02	0.33
Cu_relarea_top	0.18	0.20	0.00
Cu_relrgr_auc	-0.06	-0.05	0.82
Pb_a_auc	0.04	0.05	0.16
Pb_da_top	0.03	0.07	0.10
Pb_logIC50	0.02	0.04	0.20
Pb_relarea_auc	-0.02	0.00	0.44
Pb_relarea_top	-0.00	0.08	0.09
Pb_relrgr_auc	-0.03	-0.05	0.83
d0_a	0.23	0.23	0.00
d0_lnA	0.24	0.24	0.00
d0_rgr	0.10	0.10	0.05

Table 9. Intraclass correlation of strain-level traits by lineage (one-way ANOVA estimator, 14 lineages), before and after removing run effects (fixed run effects, lineage as a random group).

- Lineage explains up to 29% of a trait (Cr relative area). With 14 lineages each ICC has a wide interval. This is a broad-sense measure: it mixes additive and non-additive genetic effects and anything else that differs between lineages.
- Pb traits and the growth-rate traits show no lineage component (ICC at most 0.10).

5. GWAS

Three scans per trait (GEMMA, univariate LMM, Wald test, centred relatedness matrix from all 244,956 SNPs):

- (a) relatedness only, unadjusted phenotypes;
- (b) relatedness, phenotypes with run effects removed;
- (c) as (b) plus lineage indicator covariates, so that only variation inside lineages is tested.

Bonferroni thresholds use the number of distinct genotype patterns (7,199 for (a) and (b), $p < 6.95\text{e-}6$; 2,940 for (c), $p < 1.7\text{e-}5$). The suggestive level is 1 / that number.

trait	OLS, no relatedness	LMM (K only)	LMM, run-adjusted	LMM, run-adjusted + lineage	Bonferroni SNPs: LMM (K only)	Bonferroni SNPs: LMM, run-adjusted	Bonferroni SNPs: LMM, run-adjusted + lineage
d0_a	3.18	0.95	0.95	1.02	0	0	0
d0_lnA	1.10	1.50	1.50	1.39	0	0	0
d0_rgr	3.94	1.63	1.63	0.95	0	0	0
Cr_da_top	2.03	1.07	0.82	1.33	0	0	0
Cr_a_auc	3.31	1.24	1.09	0.74	0	0	0
Cr_relarea_top	5.37	1.03	1.29	1.04	31	0	0
Cr_relarea_auc	3.81	1.34	1.03	1.27	845	0	0
Cr_relrgr_auc	0.99	1.08	0.88	1.14	0	0	0
Cr_logIC50	3.72	1.50	1.42	1.21	0	0	0
Cu_da_top	3.21	1.06	1.04	0.31	0	0	0
Cu_a_auc	2.82	1.09	1.19	0.52	0	0	0
Cu_relarea_top	4.77	0.72	0.73	0.54	8009	10099	0

trait	OLS, no related- ness	LMM (K only)	LMM, run- adjusted	LMM, run-adjusted + lineage	Bonferroni SNPs: LMM (K only)	Bonferroni SNPs: LMM, run-adjusted	Bonferroni SNPs: LMM, run-adjusted + lineage
Cu_relaarea_top	2.78	1.10	1.27	0.75	0	0	0
Cu_relgrain_top	1.75	1	1.49	1.15	0	0	0
Pb_da_top	2.01	1.30	1.23	0.87	0	0	0
Pb_a_auc	3.80	1.06	1.10	1.12	0	0	0
Pb_relaarea_top	2.15	0.58	1.06	1.31	0	0	0
Pb_relaarea_auc	0.70	2.50	0.58	0.81	0	0	0
Pb_relgrain_top	0.44	0.75	0.34	1.27	0	0	0
Pb_logIC50	1.096	1.57	1.32	0.98	0	0	0

Table 10. Genomic inflation (λ) of the scans, and the number of SNPs below the Bonferroni threshold. “OLS, no relatedness” is the inflation of a plain regression on the run-adjusted phenotypes.

scan	loci_suggestive	loci_bonferroni	bonferroni_lineage_markers	contigs_with_bonferroni	loci_lineages_carrying_alt
unadjusted	174	64	64	15	8.00
runadj	165	44	44	9	7.00
runadj_lineage	12	0	0	0	

Table 11. Lead SNPs of the loci (SNPs within 50 kb are one locus) that are lineage markers: SNPs without variation inside any lineage.

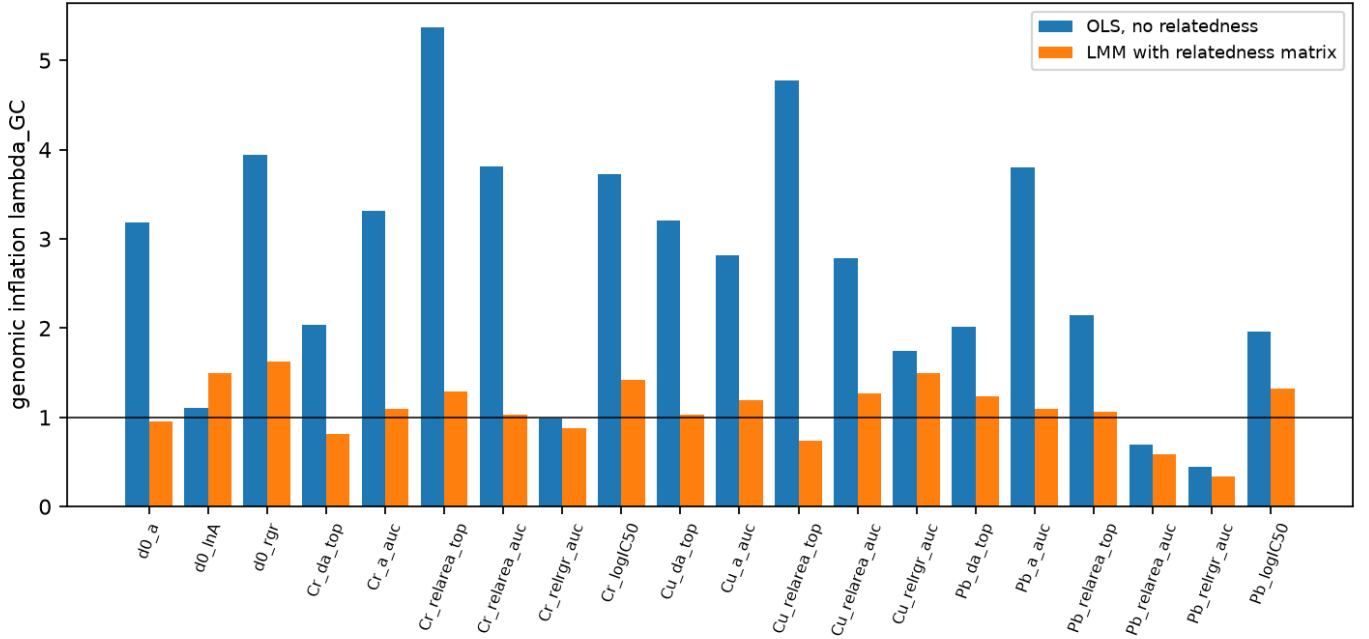


Figure 11. Inflation with and without the relatedness matrix (run-adjusted phenotypes).

Relatedness only (scans a and b)

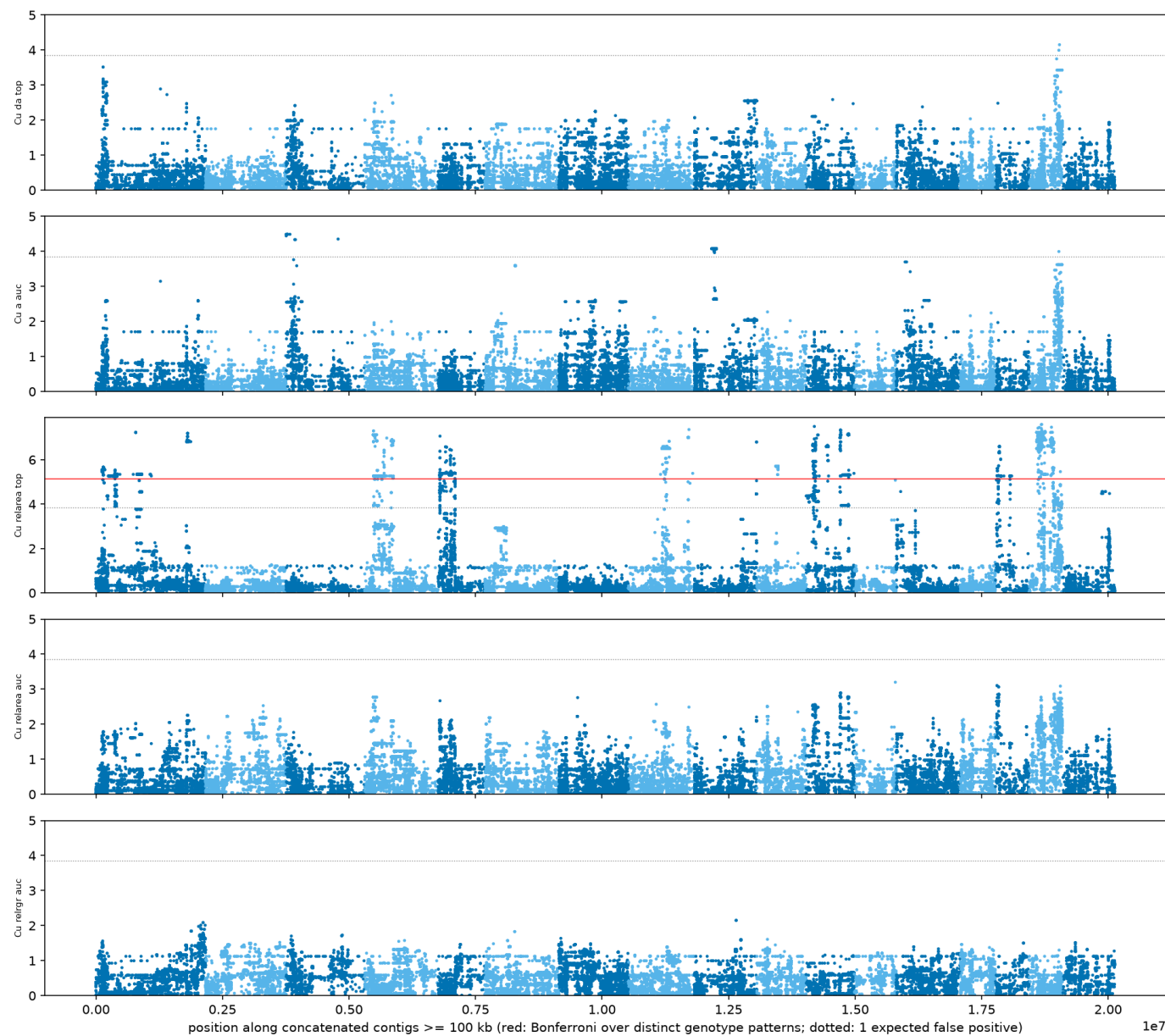


Figure 12. *Cu* traits, run-adjusted, relatedness only. The wide vertical blocks are sets of perfectly linked SNPs that mark lineages.

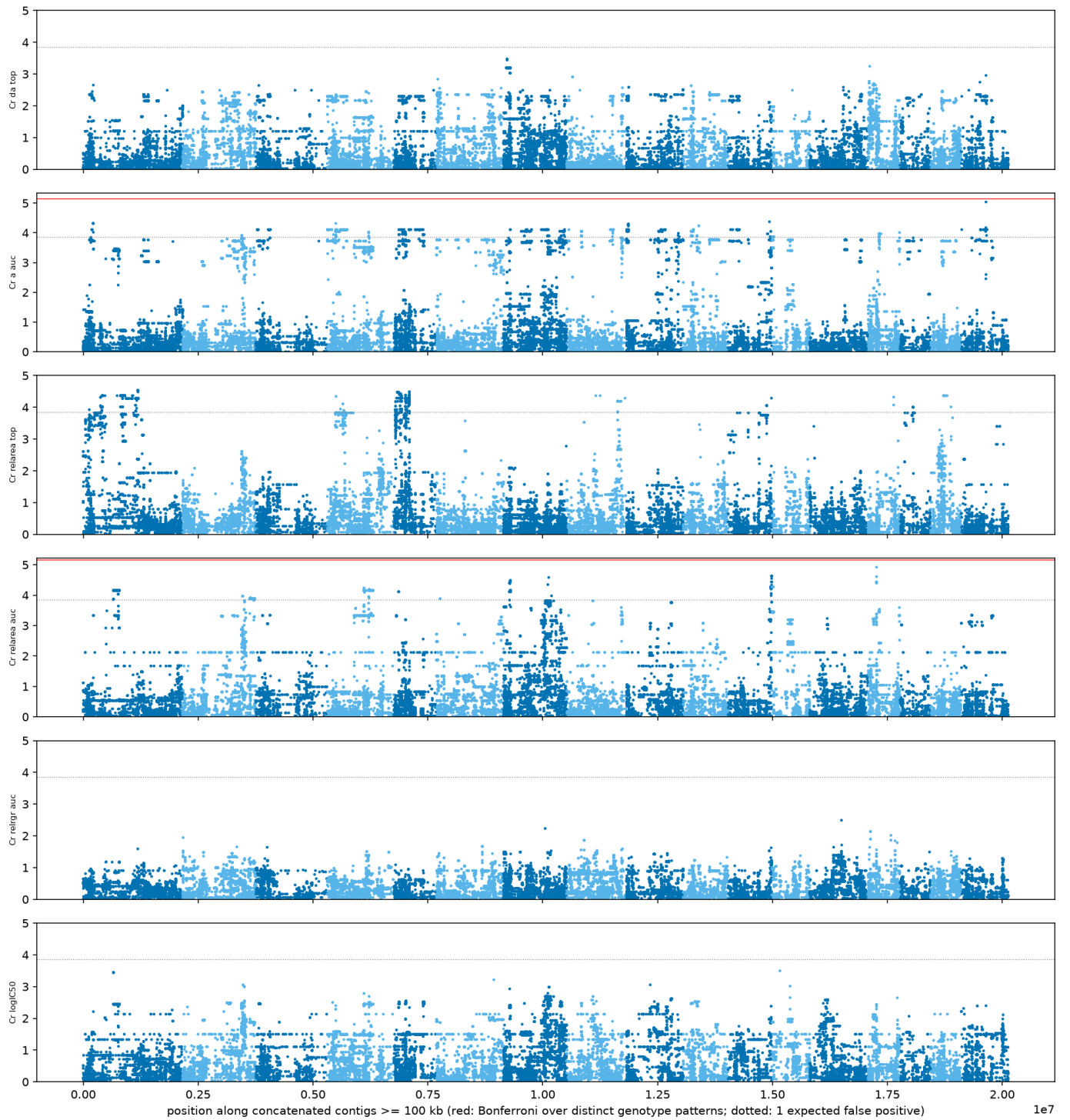


Figure 13. *Cr* traits, run-adjusted, relatedness only.

trait	chr	pos	af	beta	p_wald	suggestive	snps	in	regional	carrying	lineage	level	marked	product
Cu_relarea	CM179427	10820.192	-	4.0e-0975	938	7				True			ACY3AU_N02001	quinone reductase
Cu_relarea	CM179488	73030.225	-	5.9e-0959	379	10				True			ACY3AU_3006108	phosphinganine reductase
Cu_relarea	CM179408	07770.192	-	6.1e-0944	205	5				True			ACY3AU_hyp0528	hypoxanthine phosphotransferase
Cu_relarea	CM179487	56040.198	-	6.8e-094	448	6				True			ACY3AU_hyp0528	hypoxanthine phosphotransferase

trait	chr	pos	af	beta	p_wald	h_suggestive_snps	lineages_carrying_allele	gene	level	marked	product
Cu_relarea	CMd179491196734.208	-	7.5e- 61	0.926 08		9	True	ACY3AU_n005187			nicotetrahydrofolate reductase (NAD
Cu_relarea	CMd17949369830.199	-	7.9e- 348	0.938 08		6	True	ACY3AU_h006015			etrical protein
Cu_relarea	CMd17949870850.195	-	8.3e- 983	0.94 08		7	True	ACY3AU_n002885			PUF family protein 1
Cu_relarea	CMd17949645600.2	-	8.4e- 1215	0.928 08		7	True	ACY3AU_h002076			etrical protein
Cu_relarea	CMd17949794010.202	-	8.7e- 420	0.93 08		8	True	ACY3AU_h002021			etrical protein
Cu_relarea	CMd17949860080.204	-	8.9e- 198	0.924 08		8	True	ACY3AU_h004301			etrical protein
Cu_relarea	CMd17949366940.196	-	9.3e- 600	0.929 08		6	True	ACY3AU_i005308			ne-tRNA ligase
Cu_relarea	CMd1794988399	0.198	9.7e- 447	0.961 08		7	True	ACY3AU_h000013			etrical protein

Table 12. The 12 strongest Bonferroni loci of scan (a). Every one is a lineage marker: it carries the alternate allele in a set of whole lineages. The gene is the one at or nearest the lead SNP, and the gene is not the cause.

- A locus here is a statement about which lineages differ in the trait, and not about a gene. All SNPs on a lineage’s backbone have the same p-value. The lead SNP sits at an arbitrary position along the genome.
- The relatedness matrix does not solve this. With 14 lineages, a lineage-level trait difference is perfectly explained by thousands of SNPs.

Within lineages (scan c)

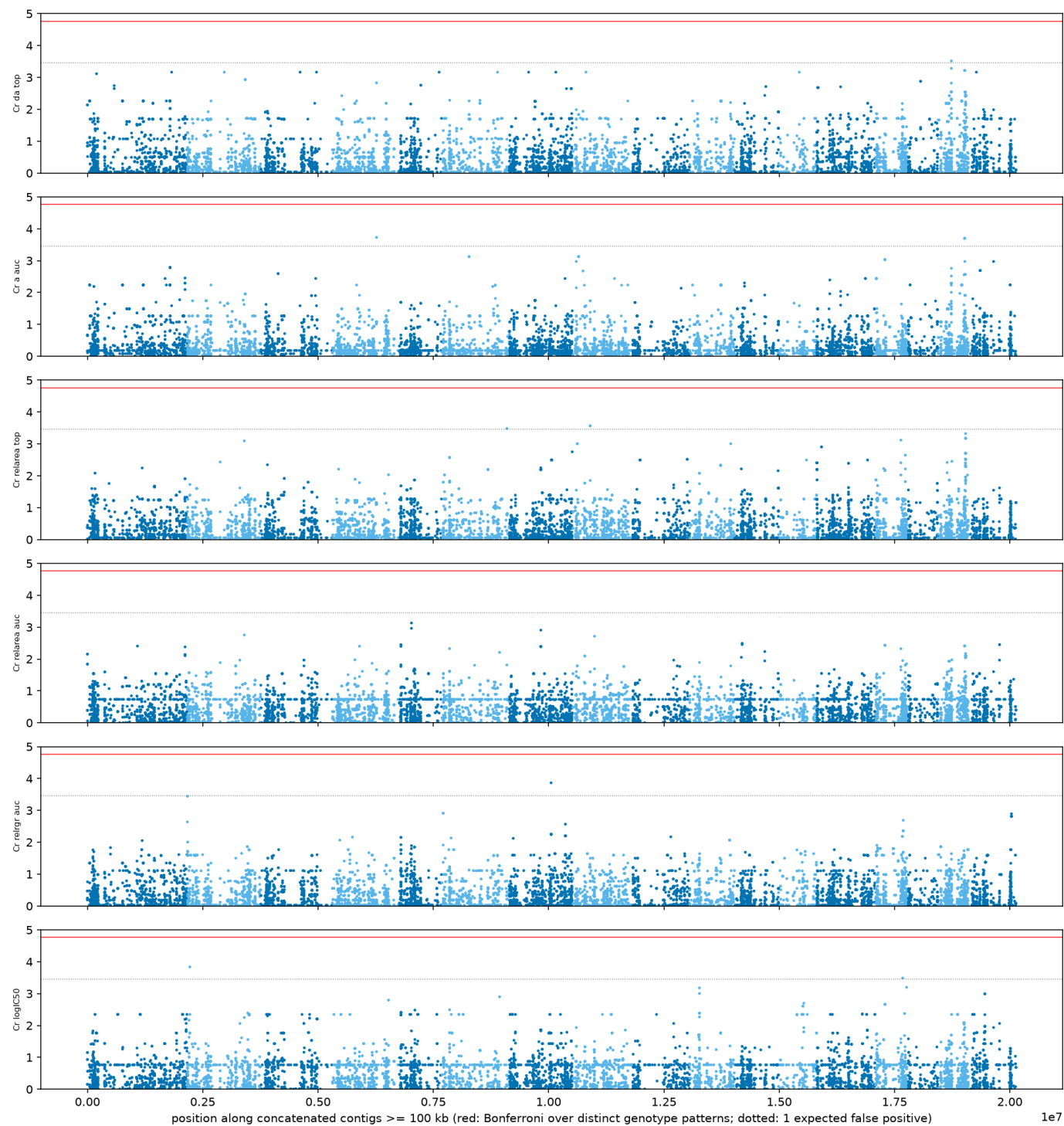


Figure 14. *Cr* traits, run-adjusted, lineage covariates.

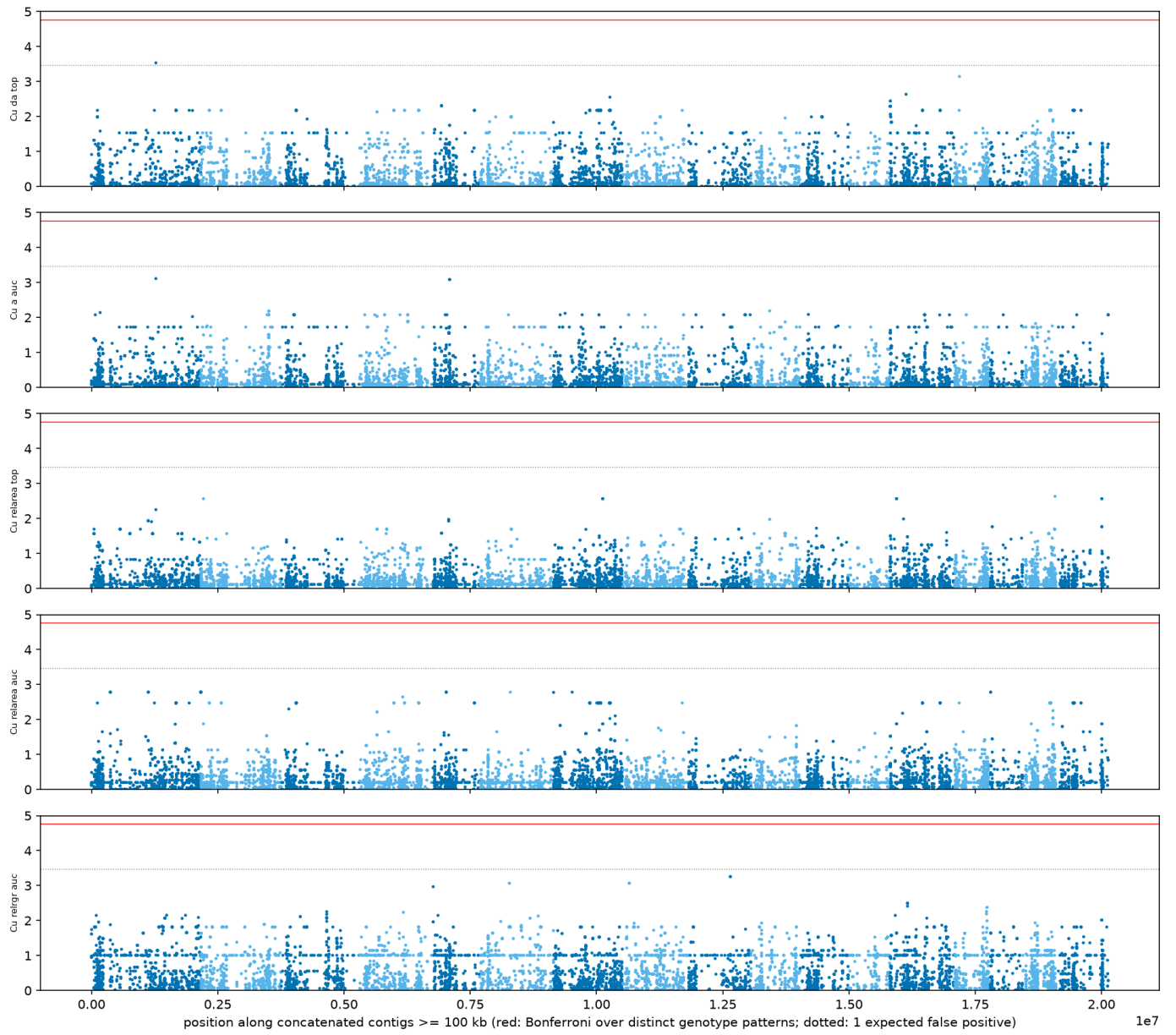


Figure 15. *Cu* traits, run-adjusted, lineage covariates.

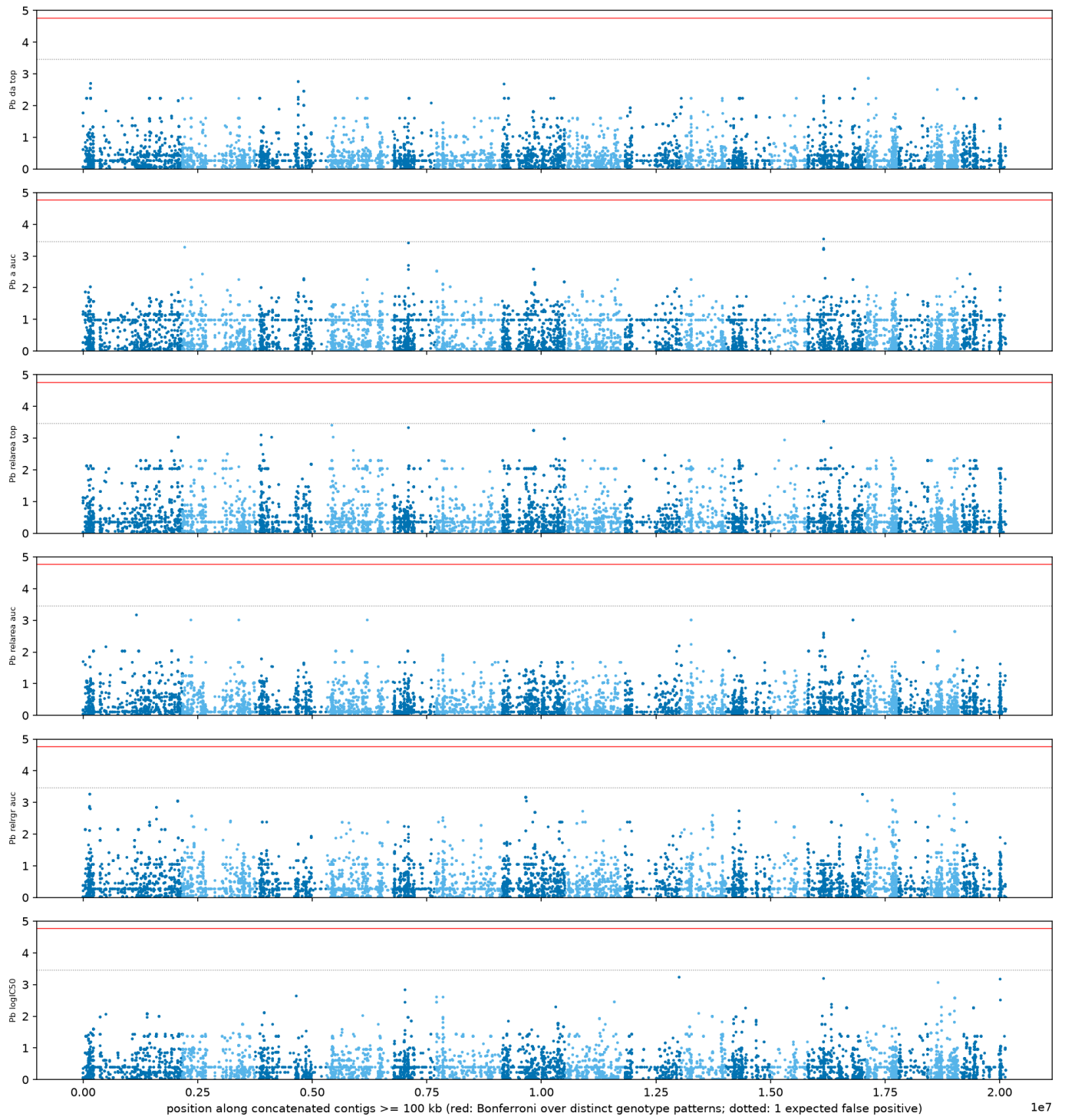


Figure 16. *Pb* traits, run-adjusted, lineage covariates.

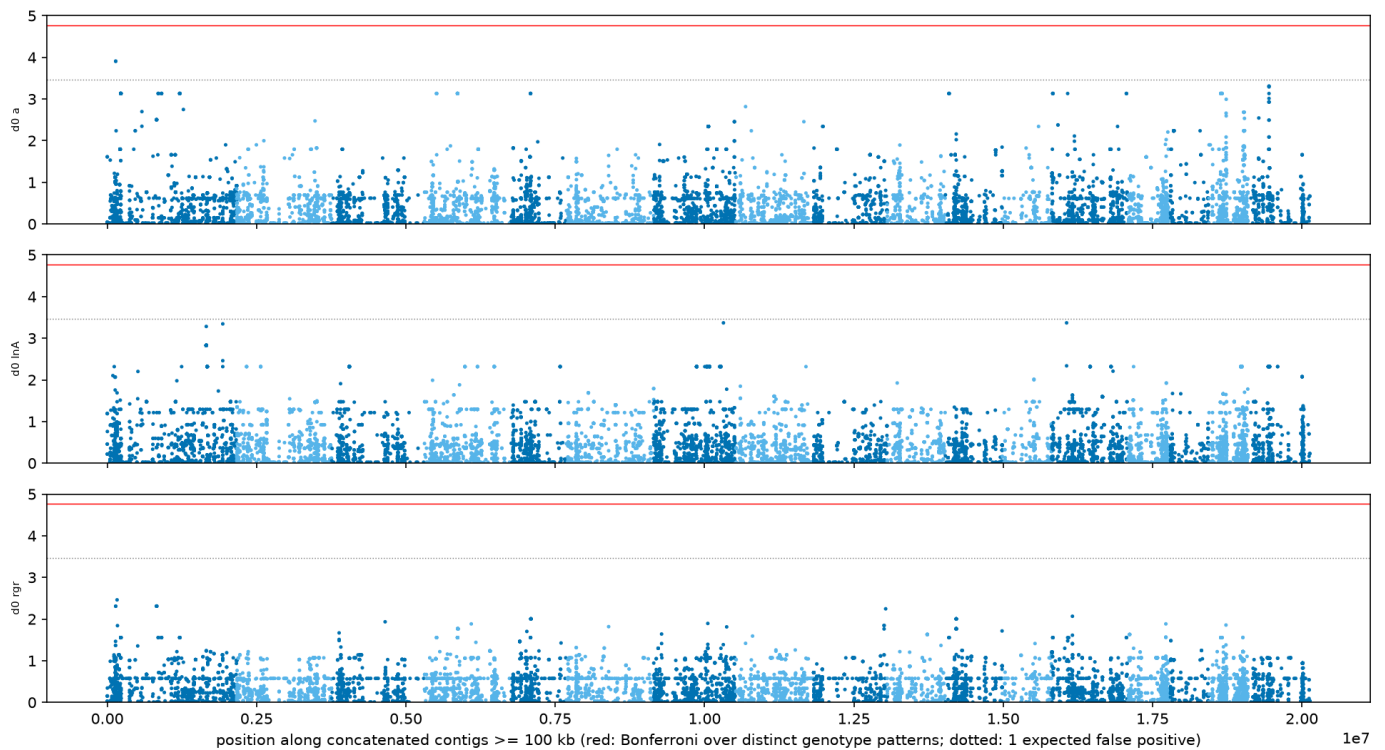


Figure 17. Dose-0 traits, run-adjusted, lineage covariates.

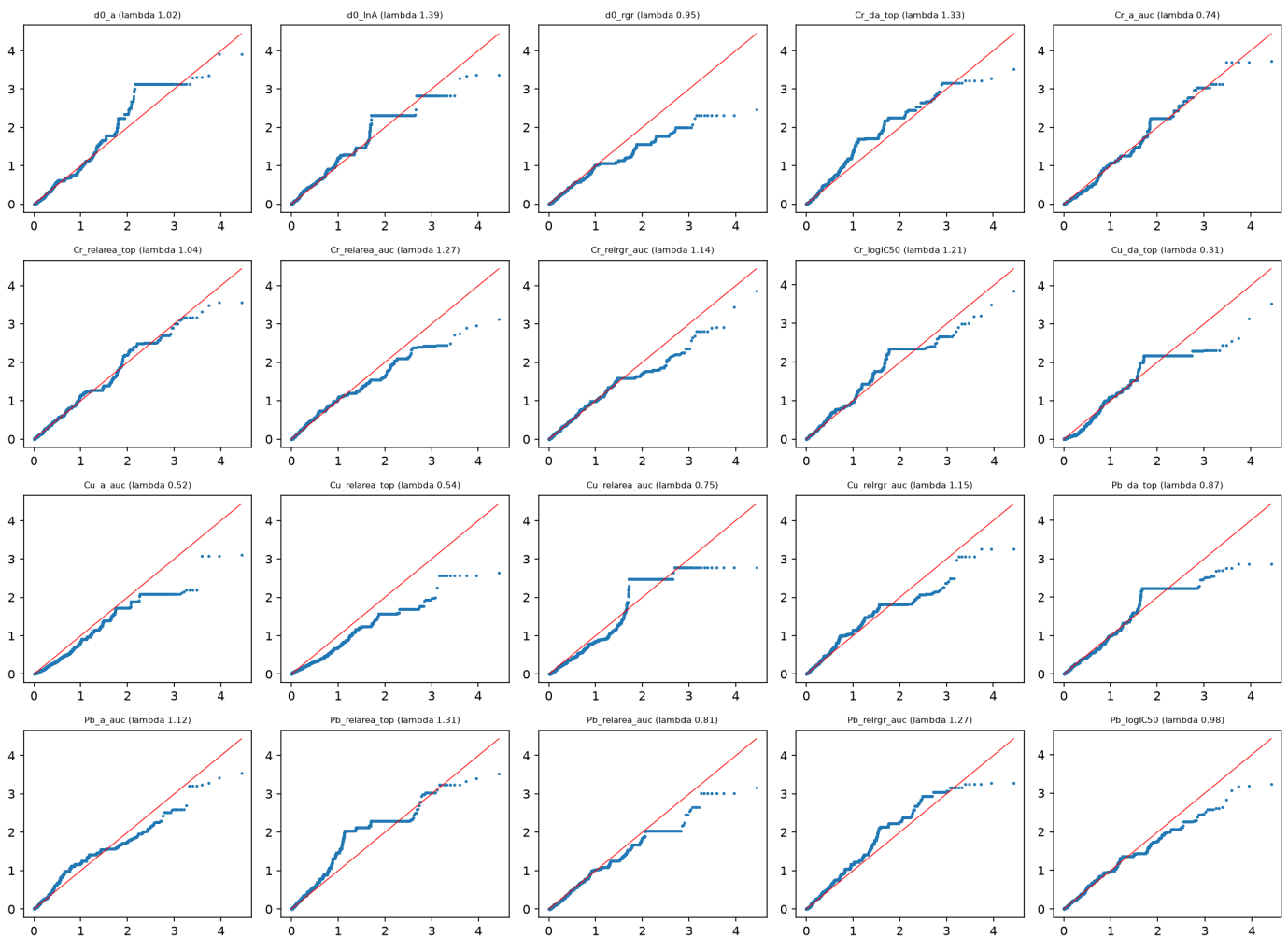


Figure 18. QQ plots of scan (c).

trait	chr	pos	af	beta	p_wald	effect	gene_id	product	distance_bp
d0_a	CM179485.1	141640	0.264	2.38	1.2e-04	upstream_gene_variant	ACY3AU_004087	cis-prenyltransferase	0
Cr_relgrain	CM179490.1	904292	0.053	-	1.4e-04	3_prime_UTR_variant	ACY3AU_005509	metabolic protein	0
Cr_logIC50	CM179486.1	55694	0.161	4.21	1.5e-04	upstream_gene_variant	ACY3AU_003323	metabolic protein	93
Cr_a_auc	CM179487.1	957384	0.168	-	1.9e-04	missense_variant	ACY3AU_005580	Nuclear control of ATPase 2	0
Cr_a_auc	CM179497.1	588055	0.211	-	2.0e-04	splice_region_variant	ACY3AU_003010	a mannosyltransferase of the dolichol pathway	0
Cr_relarea	CM179491.1	380349	0.06	-	2.7e-04	missense_variant	ACY3AU_003887	metabolic protein	0
Pb_a_auc	CM179495.1	344369	0.121	-	2.9e-04	missense_variant	ACY3AU_002286	metabolic protein	0
Cu_da_top	CM179485.1	1274990	0.132	1.63	2.9e-04	upstream_gene_variant	ACY3AU_004469	metabolic protein	20
Pb_relarea	CM179495.1	346567	0.059	-	2.9e-04	missense_variant	ACY3AU_002287	metabolic protein	0
Cr_da_top	CM179497.1	299036	0.127	1.92	3.0e-04	upstream_gene_variant	ACY3AU_003011	metabolic protein	2354
Cr_logIC50	JBZGVR010001431	5001431	0.112	-	3.2e-04	upstream_gene_variant	ACY3AU_003808	metabolic protein	102
Cr_relarea	CM179489.1	1407586	0.053	-	3.3e-04	upstream_gene_variant	ACY3AU_003443	metabolic protein	410

Table 13. Suggestive loci of scan (c) (p below $1 / 2,940$). None passes the Bonferroni threshold ($p < 1.7e-5$). *effect* is the *snpEff* effect of the lead SNP; the *gene* is the one overlapping or nearest the lead SNP, with distance in bp.

- None of the 12 loci passes the threshold, and 17 SNPs reach the suggestive level across 20 traits, against about 94 expected by chance. The scan shows no excess of signal.
- Two loci lie 2 kb apart on contig CM179495.1 (Pb a* AUC, ACY3AU_002286, and Pb relative area, ACY3AU_002287, both missense variants; $p = 2.9e-4$ each). The two traits are different summaries of the same Pb data, and the result is not replicated in an independent experiment.
- Candidates with a named product are: ACY3AU_005580 (Nuclear control of ATPase 2, Cr a* AUC, missense), ACY3AU_003010 (a mannosyltransferase of the dolichol pathway, Cr a* AUC, splice-region variant) and ACY3AU_004087 (cis-prenyltransferase, dose-0 a*). Prenyl and dolichol pathway genes are of interest for a* biology, but this is a hypothesis for later work.
- The SNP heritability estimates that GEMMA reports with lineage covariates are at the boundary (0 or 1) and are not used.

6. Non-additive architecture

Strains are haploid, so there is no dominance. Non-additive variance here means epistasis. Two analyses:

- **Variance components (REML).** Model: additive kernel (centred relatedness) plus additive-by-additive kernel (its element-wise square) plus error. The epistatic component is tested by a likelihood ratio test with the boundary correction.
- **Pairwise scan.** All pairs of the 1,401 LD-pruned SNPs ($r^2 < 0.5$), interaction term in an LMM that is adjusted for relatedness. Pairs need at least 5 strains in each of the four two-locus classes. Between 51,888 and 65,081 of 980,700 pairs per trait meet this (the others are collinear or rare because of the lineage structure).

trait	n	h2_additive_only	h2_additive	h2_epistatic_AA	residual_share	LRT	p_epistatic
d0_a	125	0.264	0.000	0.216	0.784	1.711	0.095

trait	n	h2_additive_only	h2_additive	h2_epistatic_AA	residual_share	LRT	p_epistatic
d0_lnA	125	0.000	0.000	0.160	0.840	0.000	1.000
d0_rgr	125	0.034	0.034	0.000	0.966	0.000	0.500
Cr_da_top	125	0.091	0.056	0.037	0.907	0.191	0.331
Cr_a_auc	125	0.184	0.177	0.007	0.816	0.006	0.470
Cr_relarea_top	125	0.155	0.155	0.000	0.845	0.000	0.500
Cr_relarea_auc	125	0.129	0.123	0.006	0.871	0.003	0.479
Cr_relrgr_auc	124	0.058	0.000	0.062	0.938	0.893	0.172
Cr_logIC50	123	0.045	0.045	0.000	0.955	0.000	0.500
Cu_da_top	125	0.177	0.173	0.004	0.824	0.001	0.489
Cu_a_auc	125	0.228	0.228	0.000	0.772	0.000	0.500
Cu_relarea_top	125	0.162	0.162	0.000	0.838	0.000	0.500
Cu_relarea_auc	125	0.065	0.027	0.043	0.930	0.149	0.350
Cu_relrgr_auc	125	0.000	0.000	0.012	0.988	0.311	0.289
Pb_da_top	123	0.087	0.012	0.144	0.844	0.963	0.163
Pb_a_auc	123	0.043	0.000	0.051	0.949	1.243	0.132
Pb_relarea_top	123	0.000	0.000	0.000	1.000	0.000	1.000
Pb_relarea_auc	123	0.006	0.006	0.000	0.994	0.000	0.500
Pb_relrgr_auc	122	0.000	0.000	0.000	1.000	0.000	1.000
Pb_logIC50	119	0.060	0.060	0.000	0.940	0.000	0.500

Table 14. Variance shares from the REML models (RINT traits, panel strains). **h2_additive_only** is the SNP heritability with the additive kernel alone.

trait	n_pairs_tested	bonferroni_threshold	min_p	lambda_gc	n_bonferroni	n_expected_at_1e_5	n_at_1e_5
d0_a	65081	7.68e-07	3.1e-04	0.909	0	0.651	0
d0_lnA	65081	7.68e-07	1.3e-04	0.986	0	0.651	0
d0_rgr	65081	7.68e-07	4.0e-04	0.63	0	0.651	0
Cr_da_top	65081	7.68e-07	0.001	0.952	0	0.651	0
Cr_a_auc	65081	7.68e-07	0.002	1.17	0	0.651	0
Cr_relarea_top	65081	7.68e-07	1.9e-04	0.88	0	0.651	0
Cr_relarea_auc	65081	7.68e-07	7.4e-05	1.17	0	0.651	0
Cr_relrgr_auc	65081	7.68e-07	8.5e-04	1.73	0	0.651	0
Cr_logIC50	62079	8.05e-07	7.3e-05	1.11	0	0.621	0
Cu_da_top	65081	7.68e-07	6.1e-04	1.06	0	0.651	0
Cu_a_auc	65081	7.68e-07	1.1e-04	0.755	0	0.651	0
Cu_relarea_top	65081	7.68e-07	8.2e-04	0.525	0	0.651	0
Cu_relarea_auc	65081	7.68e-07	4.9e-05	1.04	0	0.651	0
Cu_relrgr_auc	65081	7.68e-07	6.6e-04	1.56	0	0.651	0
Pb_da_top	64402	7.76e-07	2.3e-04	1.1	0	0.644	0
Pb_a_auc	64402	7.76e-07	1.5e-04	1.21	0	0.644	0
Pb_relarea_top	64402	7.76e-07	0.003	1.46	0	0.644	0

trait	n_pairs_tested	bonferroni_threshold	min_p	lambda_gc	n_bonferroni	n_expected_at_1e_5	n_at_1e_5
Pb_relarea_auc	64402	7.76e-07	0.002	0.768	0	0.644	0
Pb_relrgr_auc	64402	7.76e-07	0.002	0.596	0	0.644	0
Pb_logIC50	51888	9.64e-07	8.0e-05	0.99	0	0.519	0

Table 15. Pairwise interaction scan. No pair passes the Bonferroni threshold. **n_expected_at_1e_5** is the number expected below 1e-5 by chance.

- SNP heritability from the additive kernel alone is 0 to 0.26: d0 a* 0.26, Cu a* AUC 0.23, Cr a* AUC 0.18, Cu a* change 0.18, Cr and Cu relative area at the top dose 0.16. Standard errors are 0.08 to 0.17 (GEMMA, `gwas_inflation_pve_unadjusted.csv`).
- The epistatic component is zero or small in all traits. The smallest p is 0.095 (d0 a*). When it is non-zero it takes variance away from the additive component (d0 a*: 0.26 additive only, 0.00 additive and 0.22 epistatic). The two kernels are highly correlated in a clonal panel, so the split between them is not identifiable.
- The pairwise scan has no hit and no excess (lambda 0.5 to 1.7; median 1.02).
- With 14 lineages, no analysis can tell an additive lineage effect from an epistatic one. Both are a difference between whole genotypes.

7. Limits and next steps

- **The panel cannot support gene mapping.** Every genome-wide signal tracks lineage. A usable panel needs many more independent lineages or recombinant strains (crosses, or populations with sexual recombination). D-49 (5-SNP cutoff) should be revisited: at 2,000 SNPs only 14 strains remain.
- **Run is confounded with lineage** (all screens). Run adjustment is therefore partial. A design with lineage spread over runs, or repeated runs, would separate them.
- **Replicate noise.** The strain share of the dose-0 variance is 38% for a* and lower for size and growth. Strain-level traits from 1 to 4 wells per dose are noisy, which lowers power further.
- **Cu IC50** is censored for 46% of panel strains; only coarse.
- **Fe and Zn** are held out (D-56). When complete tables exist, add them to METALS in `01_phenotypes.py`.
- **Population analyses** (DH4148 SNP clusters, D-51) are not part of this report. The lineage table (`results/lineages.csv`) is a first version of that grouping.
- **Annotation.** Gene products come from the assembly GFF only. A GO and KEGG annotation of DH4148 would help to interpret candidates.
- **Scope of claims.** Candidate genes in section 5 are suggestive only. They have not been replicated.